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# A maize-centric framework for explainable artificial intelligence in decoding drought tolerance mechanisms

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## Abstract

Climate change-induced drought threatens global food security, with reports indicating that maize yield losses can exceed 30% in vulnerable regions, such as sub-Saharan Africa and South Asia. While traditional approaches struggle to connect genomic insights with field-level solutions, explainable artificial intelligence holds promise for transforming drought-resilience research. This review synthesizes the literature to present a conceptual, maize-centric framework for integrating explainable artificial intelligence across biological scales. We discuss how explainable artificial intelligence techniques, such as Shapley Additive Explanations and attention mechanisms, are being used to identify key drought-response elements, from novel promoter motifs to ABA signaling genes. The integration of these predictions with CRISPR screening is highlighted as a promising validation pathway, demonstrating cross-species applicability. In phenomics, we examine how unmanned aerial vehicle-based hyperspectral imaging, paired with explainable artificial intelligence, achieves high stress classification accuracy in maize, with similar protocols being developed for other cereals. For environmental monitoring, we review how models combining Internet of Things sensors with Long Short-Term Memory networks improve drought forecasts. Furthermore, architectures like the Stress-Detection Network are presented as a template for multi-crop integration. The review also addresses ongoing challenges, including data bias and computational barriers, and explores potential solutions such as federated learning and edge computing. Collectively, the literature suggests that this framework could help accelerate breeding cycles and stabilize yields under drought. These advances position explainable artificial intelligence as a pivotal tool, with maize serving as a model to bridge discovery and application for next-generation crop improvement.

**Keywords** Drought resilience, Precision phenotyping, Multi-omics integration, CRISPR-validated genes, Edge computing

## 1 Introduction

Maize (*Zea mays* L.) serves as an excellent model system for applying explainable artificial intelligence (XAI) to decode drought tolerance. Its suitability is driven by its genomic complexity, extensive phenomic datasets, and critical role in global food security [1, 2]. With yield losses exceeding 30% in vulnerable regions, developing drought-resilient varieties has become an urgent priority [3, 4]. These losses are most severe in maize-dependent regions such as sub-Saharan Africa and South Asia, where climate change exacerbates the frequency and intensity of drought [5]. Although pioneering XAI applications in crops like wheat and rice have demonstrated interpretability, they often face challenges in field validation and transferability across diverse agroecosystems. This highlights a critical gap that our conceptual maize-centric framework aims to address by leveraging maize's well-characterized genetic and phenotypic diversity [6].

The maize-focused XAI framework presented in this review offers insights that are transferable to other crops facing similar drought challenges. Despite decades of research that have elucidated fundamental drought response mechanisms, from ABA-mediated stomatal regulation to oxidative stress mitigation and epigenetic memory, translating these discoveries into improved crop varieties remains difficult [7, 8]. Conventional breeding approaches often struggle to account for three key complexities critical for drought resilience in maize and other crops: (1) nonlinear interactions between adaptive traits, such as the trade-off between water-use efficiency and nitrogen use efficiency observed in wheat [9]; (2) temporal dynamics of stress responses, exemplified in soybean, where distinct genetic modules govern drought tolerance at vegetative versus reproductive stages [10]; and (3) context-dependent genotype × environment × management (G×E×M) interactions, as seen in rice varieties that achieve high yield stability only under specific irrigation and planting density regimes [11–13].

Recent advances in high-throughput phenotyping technologies, from drone-based canopy imaging to single-cell transcriptomics, have produced multidimensional datasets that can address these limitations [14, 15]. Artificial intelligence, particularly deep learning, has shown considerable promise in analyzing such complex data to predict drought tolerance [16]. However, the "black-box" nature of conventional AI obscures the biological reasoning behind its predictions, making them difficult to interpret and of limited practical value for guiding biological discovery or breeding decisions [17, 18]. This interpretability crisis is further compounded by the "curse of dimensionality" in plant phenomics, where many measured features may reflect noise rather than biologically meaningful signals [19, 20]. For instance, maize drought-responsive genes like *ZmEREB180* display predictive temporal expression patterns, yet their underlying mechanisms remain opaque to conventional AI approaches [8, 21].

Explainable AI offers a transformative solution to these challenges by bridging predictive power with biological interpretability through three key innovations [22] (Fig. 1). First, SHapley Additive exPlanations (SHAP) values quantify how individual genomic or phenomic features contribute to predictions, enabling causal insight for marker-assisted selection [23]. Second, attention mechanisms in transformer models can visually map regulatory hotspots in biological pathways, such as abscisic acid (ABA) signaling [24]. Third, rule-extraction algorithms distill complex models into human-interpretable decision trees, providing practical tools for breeders [25, 26].

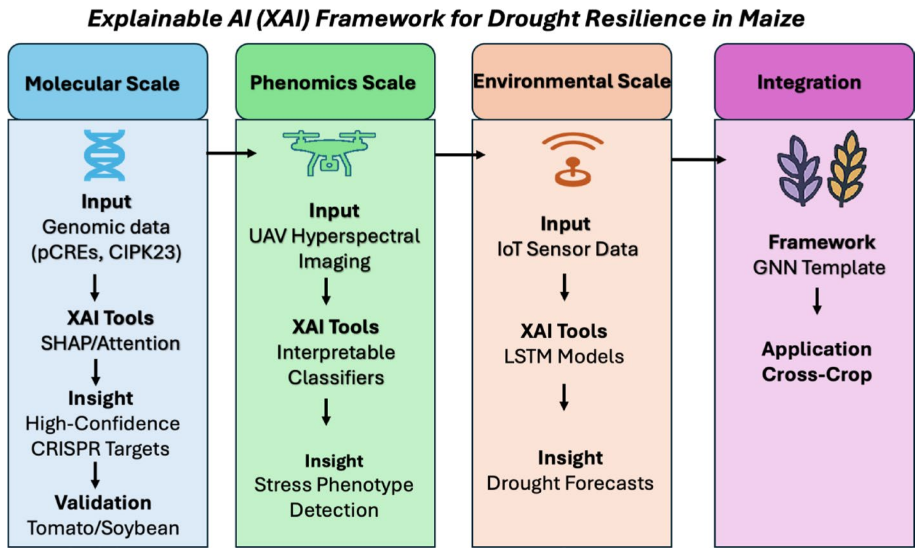


**Fig. 1** A conceptual framework for multi-scale explainable AI (XAI) in drought resilience research. The diagram illustrates a pipeline where XAI integrates data and generates biological insights across scales. At the Molecular Scale, tools like SHAP and attention mechanisms analyze genomic features (e.g., putative cis-regulatory elements/*pCREs* and genes like *CIPK23*) to identify high-confidence targets for functional validation (e.g., via CRISPR in crops like tomato and soybean). At the Phenomics Scale, interpretable classifiers analyze unmanned aerial vehicle (UAV)-based hyperspectral imaging to detect and explain stress phenotypes. At the Environmental Scale, Long Short-Term Memory (LSTM) models process Internet of Things (IoT) sensor data to generate explainable drought forecasts. The Integration phase uses architectures like Graph Neural Networks (GNNs) to synthesize these multi-scale insights, enabling knowledge transfer to other crops

These approaches show significant promise for decoding drought resilience in maize across multiple scales, from molecular and cellular mechanisms to whole-plant physiology and field-level performance, including microbial influences on root architecture and epigenetic regulation. Parallel studies in soybean and wheat suggest broader applicability, though species-specific adaptations may be required for crops with smaller genomes [23, 27, 28].

The integration of XAI into breeding pipelines has shown promising early results, with the potential to accelerate the development of drought-tolerant hybrids and improve water-use efficiency in irrigation systems [29–31]. Perhaps most exciting is the prospect of novel biological insights emerging from XAI applications, such as clarifying the roles of rhizosphere microbiome-mediated ABA signaling and transgenerational epigenetic memory. While studies on organisms like *Pseudomonas argentinensis* have demonstrated these mechanisms experimentally, XAI could offer a deeper, systems-level understanding and help uncover new strategies for drought adaptation in crops like maize [32]. Furthermore, this integration has already improved comprehension in areas such as climate-based drought prediction, where XAI models have identified key climatic drivers and elucidated their impacts on drought across different temporal scales [33]. Collectively, these advances illustrate how XAI could reshape fundamental knowledge of drought adaptation while providing more targeted, practical solutions for breeders and farmers.

Although previous reviews have extensively explored the use of AI for drought prediction [16] and stress identification in crops [34], a dedicated synthesis of XAI for decoding the biological mechanisms of drought resilience is still lacking. This review fills that gap by offering the first comprehensive overview of XAI applications in drought resilience research, using maize as a model to develop a conceptual framework adaptable to other crops. The manuscript is structured as follows: Sect. 2 outlines AI/ML workflows for multimodal drought phenotyping; Sect. 3 details the core XAI toolkit for decoding drought resilience mechanisms; Sect. 4 presents validated biological insights from XAI-driven analyses; and Sect. 5 discusses key challenges and future directions for the field. We demonstrate how the biological complexity of maize makes it an ideal testbed for XAI approaches transferable to diploid cereals and other species. We also present a methodological framework for deploying interpretable machine learning (ML) in plant phenomics, review biologically validated discoveries enabled by XAI, propose protocols for integrating these approaches into breeding programs, and suggest guidelines for implementing responsible AI in global agriculture. Although centered on maize, our conceptual framework provides a blueprint for crops facing similar drought challenges and aims to set new standards for interdisciplinary crop improvement.

## 2 AI/ML workflows for multimodal drought phenotyping

Modern drought resilience research utilizes AI and ML to integrate data from phenomics, genomics, and environmental monitoring [35]. While this multimodal approach is powerful for decoding drought responses and accelerating breeding [34] (Fig. 2), its practical value is often constrained by the "black-box" nature of complex models. This opacity obscures the biological reasoning behind predictions and limits their utility in breeding programs. This section outlines these AI/ML workflows, focusing on the interpretability challenges that necessitate the use of explainable AI (XAI).

### 2.1 High-Throughput phenomics

Remote sensing using unmanned aerial vehicles (UAVs) and hyperspectral imaging enables large-scale, non-destructive phenotyping across entire fields and breeding trials. Models such as support vector machines (SVMs) and deep neural networks (DNNs) process this data, achieving high accuracy in stress classification, often using indices like NDVI with reported accuracies exceeding 84%. Platforms such as integrated high-throughput universal phenotyping (IHUP) further automate trait extraction [36–40]. Hyperspectral data also enables genotypic ranking through methods like partial least squares regression [41]. Despite their strong predictive performance, the operational value of these AI models is significantly limited by their 'black-box' nature. This opacity obscures which specific spectral features drive predictions, reducing detections to uninterpretable correlations that provide little biological insight for breeding decisions. This interpretability gap is exacerbated by practical barriers to broader adoption, including high initial costs, substantial data storage requirements, and a lack of standardized processing protocols.

### 2.2 Genomic insights through AI

AI-driven genomic analysis bridges the gap between gene discovery and field-level resilience by integrating multi-omics data, including transcriptomics, proteomics, and epigenetics, with phenomic and environmental datasets [42]. For example, in soybean, data-driven feature engineering (DFE) has proven more effective than traditional approaches such as weighted gene co-expression network analysis (WGCNA) for identifying drought-tolerant genes in specific experimental contexts [43]. Similarly, ML

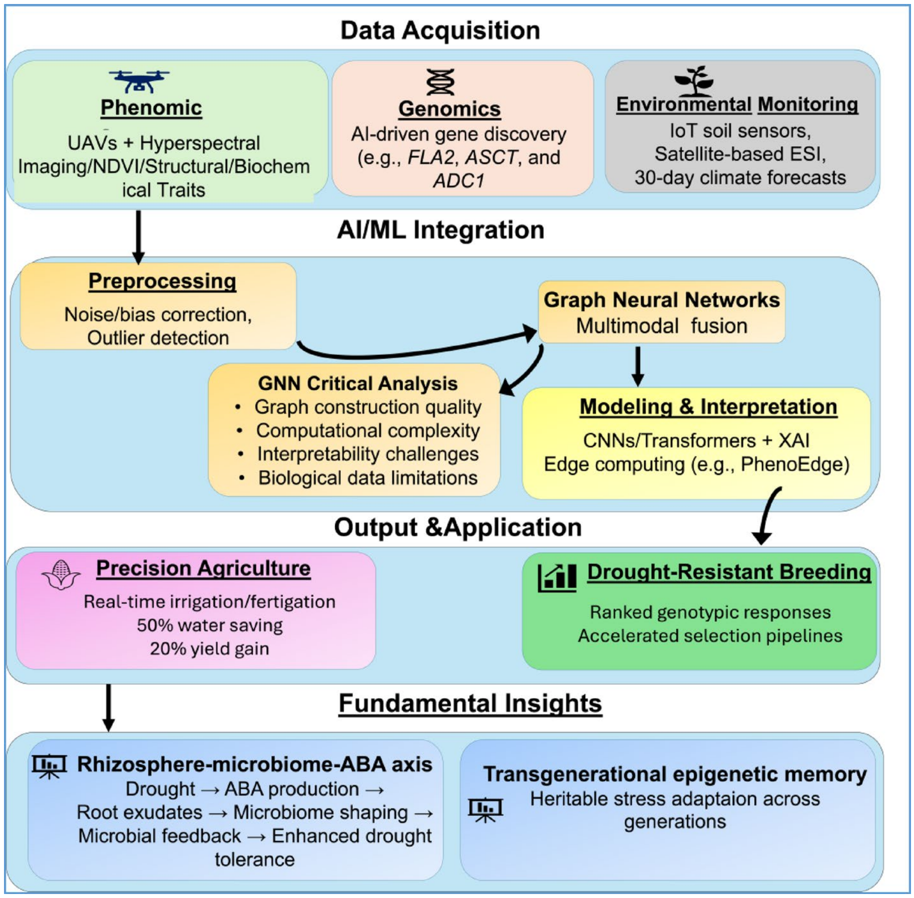


**Fig. 2** AI/ML Workflow for Multimodal Drought Phenotyping. This diagram illustrates the integrated pipeline for drought phenotyping following the acquisition-model-output narrative flow. The process begins with multimodal Data Acquisition from phenomic, genomic, and environmental sources. Data converges at the AI/ML Integration phase, where Graph Neural Networks enable multimodal fusion, with a critical analysis box highlighting methodological challenges. Processed insights drive Output & Applications in breeding and precision agriculture, ultimately yielding Fundamental Insights into the rhizosphere-microbiome-ABA axis signaling pathway and transgenerational epigenetic memory. The workflow demonstrates how AI/ML transforms raw data into actionable biological understanding and agricultural solutions

models have identified key drought-responsive genes in tomato, such as *FLA2*, *ASCT*, and *ADC1*, by mapping gene modules from RNA-Seq data integrated with phenotypic responses [44]. Beyond major cereals, AI applications extend to other crops: in strawberry, random forest models have been used to predict cultivar-specific drought effects based on traits like root length and plant height, while in wheat, AI is yielding insights into the genomic regulation of water-use efficiency (WUE) traits [45]. However, although these models excel at pattern recognition and prediction, their utility for biological discovery remains inherently constrained by their typical 'black-box' nature. This opacity limits biological interpretability, hinders the causal validation of candidate genes, and undermines the confidence required for their application in precision breeding. This fundamental limitation underscores the urgent need for XAI in genomic discovery.

### 2.3 Environmental intelligence

The integration of high-resolution environmental monitoring, such as IoT soil networks that measure volumetric water content and temperature at depths of 10–30 cm, with predictive modeling forms a critical pillar of drought resilience [46]. Hybrid models that combine physics-based simulations (e.g., HYDRUS-1D) with Long Short-Term Memory (LSTM) networks, leverage this data to provide improved 30-day drought forecasts compared to conventional indices [47, 48].

However, the operational adoption of these advanced forecasting systems is hindered by their 'black-box' nature, which obscures the rationale behind predictions and undermines user confidence, particularly among farmers. Here, XAI plays an essential role. For example, a study by Mardian, Champagne [49] used SHAP analysis to identify the key drivers within a drought model, revealing that satellite-derived evaporative stress indices (ESI) and subsoil moisture at 20–100 cm depths were more influential than atmospheric variables. Such XAI-driven insights establish a causal, interpretable link between model inputs and forecasts, moving beyond mere prediction. The translational potential is substantial; one implementation reported a reduction in water use of over 50% and a concurrent yield increase of 20% in maize through real-time irrigation management informed by such models [50]. Although these results are promising and warrant further validation, XAI effectively transforms opaque predictive systems into interpretable decision-support tools, fostering greater confidence and enabling more precise resource management.

### 2.4 Integrated workflows

Integrated AI workflows represent the frontier of multimodal data analysis in agriculture. This is exemplified by architectures like S-DNet, which fuses UAV imagery with soil moisture forecasts to achieve a 96.4% F1-score for drought recognition in winter wheat, outperforming conventional benchmarks on an independent test set [51]. Such workflows address the challenge of harmonizing data across disparate scales, from nanometer-level spectral details to kilometer-scale climate grids, often employing graph neural networks to learn cross-modal embeddings [52]. Deployment latency is mitigated through edge-computing frameworks such as PhenoEdge, enabling real-time analysis of phenotypic and environmental data streams [53].

Crucially, the integration of XAI elevates these workflows from mere prediction engines to tools for biological discovery. For example, techniques like layer-wise relevance propagation not only clarify a model's output but can also generate testable biological hypotheses by identifying the most influential input variables [54]. An XAI model might reveal, for instance, that a specific narrow hyperspectral band and root-zone soil moisture at a particular depth are the primary drivers of a drought-resilience prediction. Such insights could guide plant physiologists toward investigating the underlying mechanisms that link deep-root architecture with soil water retention.

Furthermore, these integrated AI approaches offer a powerful pathway for decoding complex biological systems. For instance, AI could analyze multi-omics and microbiome data to identify specific microbial taxa whose abundance predicts the upregulation of ABA biosynthesis genes in host plants under drought, highlighting candidate organisms for probiotic interventions [55, 56]. Similarly, ML models could mine epigenomic datasets across plant generations to detect heritable DNA methylation patterns that correlate strongly with maintained photosynthetic efficiency under water stress, thereby pinpointing candidate loci for transgenerational memory [57]. This type of integration not only accelerates breeding cycles but also holds promise for uncovering fundamental biological mechanisms, such as rhizosphere-microbiome-ABA signaling and transgenerational epigenetic memory, effectively bridging the gap between gene discovery and field-level resilience [55–59]. However, increased integration also intensifies the 'black-box' problem. Therefore, robust XAI is not merely an optional enhancement but a foundational requirement for transforming integrated agricultural AI from a powerful predictive tool into a genuine engine of biological discovery.

### 2.5 Preprocessing: handling noise, bias, and multi-modal data fusion

Robust preprocessing is essential for managing noise, bias, and multimodal data fusion in drought phenotyping workflows [60]. This stage ensures that diverse data from genomics, transcriptomics, proteomics, and metabolomics are cleaned, integrated, and optimized for accurate analysis [61]. Key techniques, such as data normalization, outlier detection, and feature selection, are employed to address data quality issues and heterogeneity [62]. Multimodal data fusion, including early fusion (combining raw data from different sources into a single input matrix) and late fusion (training separate models on different data types and merging their predictions), then enables a comprehensive view of plant drought response. This approach integrates genomic, phenomic, and environmental data, improving prediction accuracy and facilitating the identification of key drought-tolerance traits [63]. To address pervasive data imbalances, the field is increasingly moving from classical ML toward neural networks, ensemble methods, and emerging simulation-based oversampling techniques, such as Generative Adversarial Networks (GANs) [64]. Critically, the integrity of preprocessing directly dictates the reliability of XAI outputs. Noisy or biased data can produce misleading explanations; therefore, rigorous preprocessing is a prerequisite for biologically trustworthy XAI results that reflect true biological signals rather than data artifacts.

### 2.6 Model selection: black-box vs. interpretable models

The selection of AI models for drought phenotyping is guided by key criteria, including dataset size and dimensionality, the required predictive performance, and, crucially, the need for biological interpretability. This often creates a trade-off between predictive power and model transparency [65]. Black-box models, such as Convolutional Neural Networks (CNNs) and transformers, excel with large, complex datasets like hyperspectral imagery, capturing subtle patterns for accurate stress detection [66, 67]. However, their opacity obscures the decision-making process, and their high complexity increases susceptibility to overfitting on noisy or limited datasets, presenting a dual challenge for biological applications [54, 68].

In contrast, inherently interpretable models such as decision trees and Bayesian networks provide direct transparency into feature contributions, offering immediate insight into biological mechanisms [69, 70]. Ensemble methods like random forests strike a balance, often delivering robust performance with built-in feature importance metrics, though they may not achieve the peak accuracy of deep learning models on highly complex tasks [71]. Given these trade-offs, a hybrid approach is frequently the most practical solution. Applying post-hoc XAI techniques like SHAP to powerful black-box models can yield high predictive accuracy while also revealing the reasoning behind predictions [33, 72]. This strategy directly addresses the core limitations of complex AI, transforming opaque outputs into transparent, biologically grounded insights for researchers and breeders.

## 3 XAI toolkit for drought resilience

The opacity of conventional machine learning models has significantly hindered progress in drought resilience research, particularly in maize, where complex genotype-phenotype-environment interactions require high interpretability [65]. Explainable AI techniques are now illuminating these black-box models, transforming predictive outputs into mechanistic insights across genomic, phenomic, and environmental domains [73, 74] (Fig. 3). This section introduces and compares the core XAI methods that form the essential toolkit for discovery in agricultural science.

### 3.1 Feature attribution methods

Feature attribution techniques quantify the contribution of individual input features to model predictions. Among these, SHAP has gained significant traction for assigning importance scores in diverse agricultural models [75]. A critical distinction exists between local and global interpretability: SHAP delivers global explanations by aggregating local feature importance across all predictions, offering a comprehensive view of

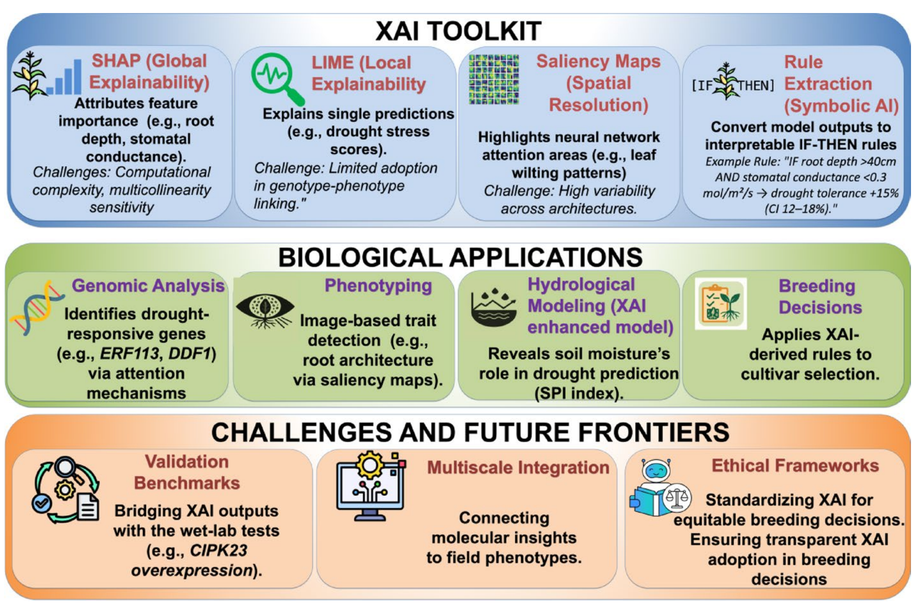


**Fig. 3** Conceptual Explainable AI Framework for Decoding Drought Resilience Mechanisms. This diagram illustrates a comprehensive pipeline that integrates machine learning interpretability with agricultural bioscience. The framework is structured around three core components: (1) The XAI Toolkit showcases four interpretability methods, including SHAP for global feature importance analysis of drought traits such as root depth and stomatal conductance [75], LIME for local prediction explanations [78], saliency maps for visualizing neural network attention in image-based phenotyping [79], and rule extraction for generating biologically actionable IF-THEN decision rules [85]. (2) Biological Applications demonstrate how these XAI outputs drive genomic analysis, for instance, identifying key drought-responsive genes such as *ERF113* and *DDF1* through attention mechanisms [90], enable image-based phenotyping of root architecture, enhance hydrological modeling of soil moisture via the SPI index [91], and inform breeding decisions [76]. (3) Implementation Challenges highlight critical hurdles, including the biological validation of XAI predictions (e.g., *CIPK23* wet-lab verification [76], multiscale integration from molecular to field-level insights [86], and the development of ethical frameworks for responsible deployment in breeding programs [87]. Collectively, the figure shows how these interconnected components transform opaque machine learning models into transparent, biologically grounded tools for drought resilience research, creating a closed loop from computational analysis to actionable agricultural insights while acknowledging current limitations in validation and scalability

feature relationships. In contrast, LIME focuses exclusively on explaining individual predictions locally. In practice, SHAP feature sets are highly context-dependent. Genomic applications often use k-mer frequencies, epigenetic marks, and known transcription factor (TF) binding sites, whereas phenomic analyses rely on spectral indices, morphological traits, and environmental covariates [76]. Although SHAP's direct application to identifying traits like root system architecture or specific gene-gene interactions in drought phenotyping remains limited, it holds strong promise for improving model transparency and guiding biological hypothesis generation [88, 89].

However, SHAP faces several limitations beyond its high computational complexity and sensitivity to multicollinearity. These include feature dependence issues, which can distort importance rankings in biological datasets; instability across different background data distributions; and difficulties in handling the high-dimensional, correlated features common in genomic and phenomic data. Addressing these limitations requires careful implementation, often involving model-specific approximations (e.g., TreeSHAP for tree-based models) or the analysis of representative data subsets rather than full datasets [77]. For drought phenotyping applications, SHAP is generally preferable for identifying features that are consistently important across genotypes, whereas LIME is more effective for explaining model behavior under specific environmental conditions or for rare genetic variants. While SHAP offers global insights, LIME provides local, linear approximations for individual predictions and has been applied in spectral data analysis, though it remains less common in plant phenotyping [78]. LIME's primary strength is its ability to deliver intuitive, instance-specific explanations, even though it may lack the global consistency of SHAP.

### 3.2 Visual and structural interpretation methods

Visual methods provide intuitive insights into model focus areas. Saliency maps, for instance, are commonly used to visualize neural network attention in image-based phenotyping. However, benchmark studies reveal substantial variability in saliency maps across different network architectures, highlighting the need for further methodological refinement and standardization [79]. In contrast, attention mechanisms in transformer models have transformed sequence-based analysis by dynamically weighting the importance of specific input elements [80]. For genomic applications, these models typically define 'tokens' as fixed-length nucleotide sequences (k-mers) or individual genes, which are normalized for sequence length and composition before being embedded into a continuous vector space [80]. These biologically informed architectures can uncover functional networks by analyzing their learned attention patterns. This approach captures more than simple feature importance, revealing complex non-linear relationships, as illustrated in hydrological modeling, where XAI techniques have elucidated the role of soil moisture in streamflow prediction [81]. Furthermore, integrating attention mechanisms with rule extraction systems can enhance explainability by both highlighting influential inputs and generating interpretable decision rules [82].

### 3.3 Rule-Based explanation systems

Rule extraction methods translate complex model decisions into human-interpretable logic. Symbolic AI approaches, such as decision trees and Logic Learning Machines (LLMs), distill intricate patterns into actionable "if-then" rules [83, 84]. These methods can, in principle, yield direct insights into drought resilience mechanisms. Decision trees, valued for their inherent transparency, have been shown to reduce the time breeders require for interpretation compared to conventional GWAS outputs, offering a more accessible link between AI-driven predictions and practical breeding decisions [85].

### 3.4 Implementation challenges and future directions

Looking ahead, three critical frontiers will define the future role of XAI in drought resilience research. First, establishing robust biological validation benchmarks is essential to anchor XAI interpretations in empirical evidence and ensure their reliability in real-world applications [73, 76]. Second, multiscale integration frameworks are needed to connect molecular-scale explanations with field-level phenotypes, enabling the practical translation of XAI insights across stages of crop development [86]. Third, ethical and governance frameworks must be developed to guide the responsible use of XAI in breeding decisions, ensuring transparency, fairness, and accountability for technologies that directly affect global food security [59, 87]. Collectively, this evolving toolkit is advancing drought phenotyping from correlative observation toward mechanistic understanding. The integration of global feature attribution (SHAP), local interpretability (LIME), spatial insights from saliency maps, and the contextual focus of attention mechanisms offers a multidimensional analytical lens. Together, they are reshaping AI from a predictive instrument into a discovery engine for fundamental plant biology [76, 80, 88, 89].

While powerful, these XAI methodologies must be applied with a clear understanding of their limitations and the data quality challenges inherent in multimodal drought phenotyping. As discussed in Sect. 5, issues such as dataset imbalance, annotation inconsistency, and environmental noise directly affect the reliability of XAI-derived interpretations, underscoring the necessity of rigorous biological validation.

## 4 Decoding drought mechanisms via XAI: validated biological insights

Recent advances in applying the XAI toolkit have yielded profound, experimentally validated insights into the molecular mechanisms of drought tolerance (Fig. 4). Through the integration of multi-omics data and XAI techniques, researchers are now deciphering complex regulatory networks, linking these networks to key drought-response pathways, and uncovering valuable genetic targets for crop improvement.

### 4.1 Elucidating cis-regulatory logic in rice

A seminal study applied XAI to achieve a mechanistic breakdown of gene regulatory logic under drought stress in rice [92]. To ensure model robustness and mitigate overfitting, a critical concern with high-dimensional genomic data, the researchers implemented a rigorous validation framework. This included a gene-family-guided train-test split and nested cross-validation. They trained Random Forest models on a comprehensive set of genomic features, which comprised: (i) nucleotide and dinucleotide composition across promoter, coding, and untranslated regions; (ii) known transcription factor binding sites (TFBSs) from established databases; and (iii) novel putative cis-regulatory elements (pCREs), identified as enriched 6–8-mer oligonucleotides within drought-responsive promoters.

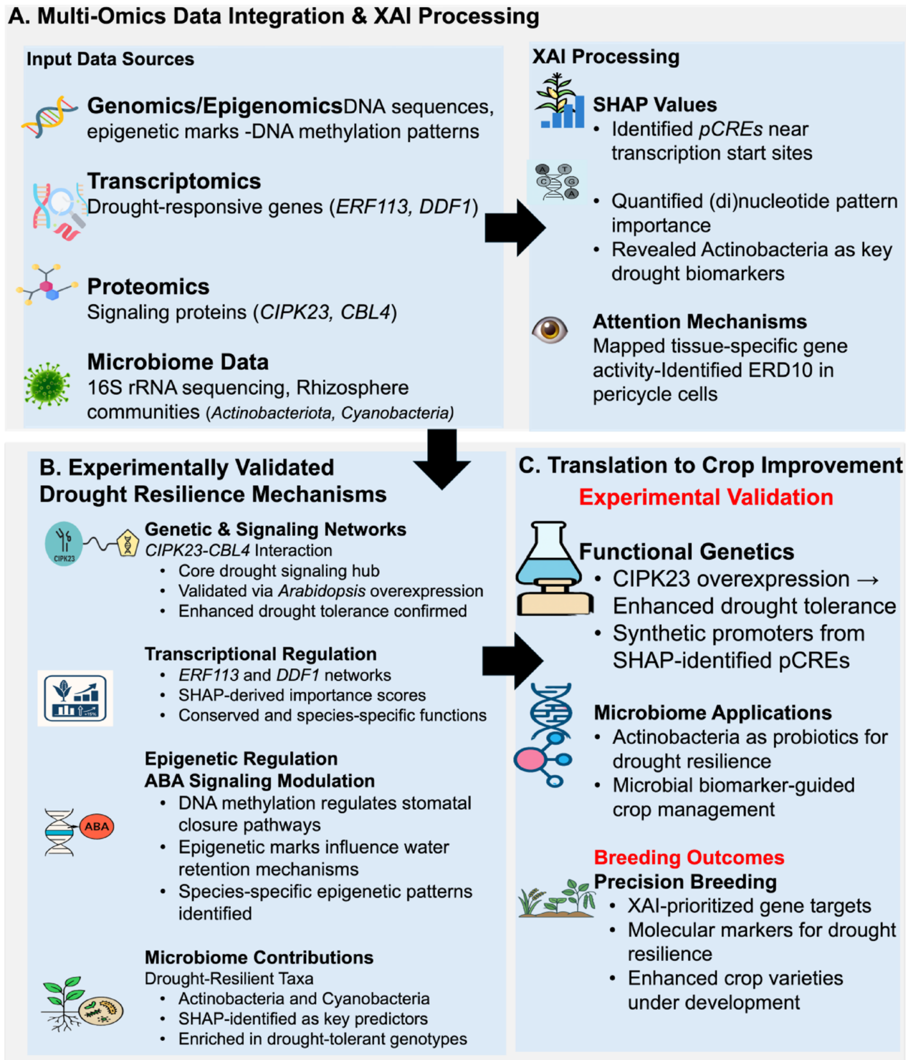


**Fig. 4** An XAI-guided conceptual framework for decoding drought resilience mechanisms through multi-omics integration. The framework synthesizes experimentally validated findings from published studies, showing the pipeline from (**A**) multi-omics data integration and XAI processing to (**B**) biologically validated mechanisms and (**C**) translational applications. Key components are supported by specific research: SHAP analysis revealed *pCREs* and (di)nucleotide patterns as key genomic predictors [92], while microbiome analysis identified Actinobacteria and Cyanobacteria as drought-resilient taxa [93]. The *CIPK23-CBL4* interaction was functionally validated through overexpression studies, and tissue-specific gene regulation was resolved through attention mechanisms [90]. Epigenetic regulation of ABA signaling pathways was demonstrated through DNA methylation patterns [32]. These discoveries enable targeted crop improvement through synthetic promoter design, microbiome management, and precision breeding

The models achieved strong predictive performance, with AU-ROC scores ranging from 0.80 to 0.86 and AU-PR scores from 0.82 to 0.87. For interpretability, SHAP was selected over other XAI tools, such as LIME, due to its solid theoretical foundation and ability to effectively capture feature interactions in tree-based models. SHAP analysis revealed that features in the proximal promoter region, within 300 bp of the transcription start site, were the most critical predictors of gene activation. A key finding was that novel *pCREs* (e.g., motifs resembling TBP, MYB, and GATA binding sites) were more predictive than known TFBSs, indicating the presence of previously unrecognized regulatory layers.

Furthermore, the dinucleotide composition of coding sequences substantially enhanced model performance when combined with *pCREs*, underscoring the functional importance of these fundamental DNA sequence elements. Biological validation was conducted through functional enrichment analysis, which linked the top-ranked features to relevant stress-response pathways. Although the identified motifs show evolutionary conservation, the direct transferability of this model to other cereals requires further validation, as regulatory architectures can be species-specific.

This discovery demonstrates cross-species relevance; for instance, a comprehensive multi-omics pipeline applied to soybean identified 10 key drought-tolerant genes, including a predicted heat shock protein (*Glyma.07g043600*) and carbohydrate metabolism genes (*Glyma.05g056300*, *Glyma.11g111400*). The pipeline employed rigorous statistical validation and qRT-PCR confirmation of gene expression under drought stress [43].

### 4.2 Discovery and validation of key signaling hubs

XAI has played a pivotal role in identifying and prioritizing core components of drought signaling networks. By integrating multi-omics approaches, XAI frameworks have demonstrated how genes such as *CIPK23*, a key protein kinase, coordinate drought responses through interactions with signaling partners like *CBL4*. The role of *CIPK23* as a central regulator has been further confirmed through experimental validation, including overexpression studies in *Arabidopsis thaliana*. The cross-species utility of XAI is also evident in studies such as one on almond, where SHAP analysis of genomic data identified regions significantly associated with shelling fraction, achieving a prediction performance of $R^2 = 0.51$ and pinpointing a candidate gene involved in seed development [76]. Moreover, XAI has helped clarify the roles of other important regulatory genes, including *ERF113* and *DDF1*, revealing how they function within both conserved and species-specific genetic networks [90].

### 4.3 Resolution of tissue and cell-type specific responses

XAI enables the analysis of drought responses with fine spatial resolution, moving beyond bulk-tissue averages. This capability has uncovered distinct, tissue-specific roles of genes such as *ERD10*, which functions differentially in cell types like the pericycle and endodermis. Revealing this cell-type-specific functionality is essential for advancing targeted breeding strategies designed to enhance drought resistance at precise developmental and anatomical levels [90].

### 4.4 Integration of epigenetic and Microbiome modulations

The integrative capacity of XAI frameworks has also shed light on the role of epigenetic modifications in drought tolerance. These analyses reveal how pathways such as ABA signaling, which governs stomatal closure and water retention, are modulated by species-specific epigenetic regulation. XAI can help optimize these pathways for more targeted genetic interventions [32]. Beyond the plant genome, XAI is increasingly used to decode plant–microbiome interactions under drought. For example, in a study of cowpea, microbial communities were profiled using 16 S rRNA amplicon sequencing. The relative abundance of bacterial taxa served as input to a Random Forest model, which achieved high predictive accuracy for drought stress (accuracy = 0.812, AUC =

0.87). SHAP analysis identified key microbial taxa as significant predictors, with the phyla Actinobacteriota and Cyanobacteria emerging as the most important biomarkers enriched under drought conditions [93]. This XAI-driven approach provided mechanistic insights into how specific rhizosphere communities influence host physiology, pointing toward new strategies for crop improvement through microbiome management. Collectively, these advances, integrating XAI with multi-omics and experimental validation, are paving the way for more efficient breeding programs. By leveraging XAI-driven insights, researchers can prioritize and validate genes and mechanisms associated with drought resistance, accelerating the development of crops resilient to increasingly challenging climatic conditions.

## 5 Challenges and limitations in AI/ML for drought resilience

Machine learning and XAI hold transformative potential for enhancing drought resilience in cereals such as maize, wheat, and rice; however, their implementation is constrained by significant technical, biological, and practical challenges. Data limitations remain a primary bottleneck. Insufficient and imbalanced datasets, especially in underrepresented regions, often lead to overfitting and poor model generalization [94]. Although advanced multimodal approaches can achieve high accuracy under controlled research conditions (e.g., 96.71% for drought identification in winter wheat using RGB and vegetation indices), such results often reflect idealized settings that may not translate reliably to field applications, where variable data quality and environmental heterogeneity pose substantial barriers [95]. Dataset imbalances can be severe. For example, in wheat phenotyping using 3D point clouds, the uneven proportion of ear to leaf points introduces pronounced intra-class imbalance, reducing segmentation accuracy. Implementing weighted sampling and tailored loss functions improved ear segmentation performance by 10–12% in mean Intersection over Union (mIoU), illustrating both the scale of the problem and the value of targeted mitigation [96]. Critically, these data limitations directly affect the applicability of our maize-centric XAI framework: models trained predominantly on temperate maize hybrids may generate misleading XAI interpretations when applied to tropical germplasm or underrepresented agro-ecologies, creating a fundamental trust barrier for breeders working across diverse environments. Noise, outliers, and spatiotemporal gaps further degrade model performance, although preprocessing techniques (e.g., wavelet transforms and variational mode decomposition), combined with remote sensing integration can partially mitigate these issues [97, 98].

Model complexity introduces a fundamental trade-off between accuracy and computational efficiency. For example, while gradient-boosted trees can outperform other algorithms in selecting microbial strains for drought mitigation, their substantial resource requirements raise scalability concerns [99]. Hybrid models that integrate ML with mechanistic approaches, such as hydraulic principles, show promise for capturing the nonlinear interactions among soil, climate, and plant variables [100, 101]. However, biases inherent in training data, such as the overrepresentation of temperate hybrids or controlled-environment conditions, can skew predictions and limit their applicability in real-world field settings [102]. These biases are compounded by environmental variability, as models trained on limited datasets often fail to generalize across diverse soil types and microclimates [103]. This mismatch creates a tangible adoption risk: breeders may receive XAI recommendations that perform well under experimental conditions but prove ineffective in specific field environments, reinforcing skepticism toward data-driven solutions. Emerging approaches such as active learning and adversarial debiasing offer promising pathways toward improving model fairness and robustness across diverse agroecosystems [104].

Environmental variability further complicates these challenges, as plant–microbe interactions and drought responses are highly context-dependent [105]. Although XAI tools like SHAP and LIME enhance interpretability by quantifying how specific biological features contribute to drought resilience, their capacity to fully supplant mechanistic models remains unclear [90, 92]. Scalability presents another hurdle: models developed in controlled environments often perform poorly under field conditions due to uncontrolled variables, highlighting the need for edge-computing solutions to enable real-time deployment [106]. Practical barriers, such as the high cost of large-scale trials, data privacy concerns, and breeder skepticism, also impede adoption [107]. For instance, breeders may distrust black-box predictions that lack biological explanations, slowing the integration of AI into crop improvement pipelines [108].

Therefore, the successful implementation of our conceptual framework requires not only technical advances but also deliberate strategies to foster domain-specific trust. These include transparent communication of model limitations, robust validation across target environments, and collaborative development with end-users to ensure XAI outputs align with practical breeding constraints. Collaborative efforts that integrate genomics, synthetic biology, and precision agriculture will be essential for bridging these gaps. Key priorities include hybrid modeling to merge ML's predictive power with mechanistic insights (e.g., coupling hydraulic models with neural networks) [109], federated learning to expand datasets while preserving privacy across institutions [110], and the development of lightweight AI tools for edge devices to support real-time drought monitoring in resource-limited settings [111]. By systematically addressing these challenges, AI and ML can evolve from theoretical promises into actionable solutions for enhancing global drought resilience.

## 6 Conclusion and future perspectives

The integration of XAI into drought tolerance research marks a transformative advance for crop improvement, bridging the long-standing gap between predictive modeling and biological insight. By leveraging interpretability techniques such as SHAP values and attention mechanisms, we can now decode complex regulatory networks, from drought-responsive promoter motifs to systemic signaling components such as *CIPK23* in ABA-mediated responses. These advances are already enabling tangible applications, including synthetic promoter design and microbiome-assisted resilience strategies, offering new pathways for modern breeding programs. However, three critical challenges must be addressed to fully realize the potential of XAI. First, harmonizing multi-omics and phenomics data across biological scales requires enhanced computational frameworks and standardized validation benchmarks. Second, although XAI has improved model transparency, field applicability remains limited by environmental variability and computational demands—challenges now being tackled through physics-informed hybrid models and edge-computing solutions. Third, integrating XAI with precision technologies such as CRISPR and real-time irrigation systems demands both technical refinement and ethical oversight to ensure responsible implementation. Looking ahead, strategic priorities should include: (1) Causal validation through integrated CRISPR-XAI platforms, where XAI prioritizes high-impact gene targets from multi-omics data and CRISPR enables rapid functional validation in planta, (2) Data equity and scalability via federated learning, expanding datasets through privacy-preserving collaborations across institutions to mitigate bias, (3) Translation and adoption by developing farmer-centric interpretability tools and delivering innovations through public–private breeding partnerships and digital extension services. Realizing this vision will require unprecedented collaboration across computational biology, breeding, and climate science. With concerted effort, these advances can significantly accelerate breeding cycles and enhance yield stability under drought. Achieving aspirational goals, such as reducing breeding cycles by 50%, will depend on the effective implementation of these strategies and the successful translation of laboratory discoveries into field-ready solutions.

**Abbreviations**

| | |
|---|---|
| AI | Artificial intelligence |
| ML | Machine learning |
| XAI | Explainable artificial intelligence |
| ABA | Abscisic acid |
| UAV | Unmanned aerial vehicle |
| NDVI | Normalized difference vegetation inde |
| SVM | Support vector machine |
| DNN | Deep neural network |
| IHUP | Integrated high-throughput universal phenotypin |
| DFE | Data-driven feature engineering |
| WGCNA | Weighted gene co-expression network analysis |
| WUE | Water use efficiency |
| IoT | Internet of things |
| LSTM | Long short-term memory, neural network |
| ESI | Evaporative stress index |
| SPI | Standard precipitation index |
| SHAP | SHapley additive explanations |
| LIME | Local interpretable model-agnostic explanations |
| CNN | Convolutional neural network |
| LLM | Logic learning machine |
| TFBS | Transcription factor binding site |
| pCRE | Putative cis-regulatory element |
| CIPK23 | CBL-Interacting protein kinase 23 |
| CBL4 | Calcineurin B-like protein 4 |
| RF | Random forest |
| GWAS | Genome-wide association studies |

**Author contributions**

Authors BQ, BAM and LAP contributed to the conception of the study, preparation, and design of this review article. The literature survey was done by the author BQ and LAP. BQ wrote the manuscript's first draft, and all authors commented on it. All authors read, revised, and approved the final manuscript.

**Funding**

Not applicable.

**Data availability**

No datasets were generated or analysed during the current study.

## Declarations

**Ethical approval**

Not applicable.

**Consent to participate**

Not applicable.

**Consent to publish**

Not applicable.

**Clinical trial registration**

Not applicable.

**Competing interests**

The authors declare no competing interests.

Received: 29 July 2025 / Accepted: 20 January 2026
Published online: 30 January 2026

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